Augmented Driver Ranking & Review System

Good day,

Thank you for taking the time to review this prototype. This document is intentionally brief. It covers the core idea, the results, and four real query examples. Your feedback will directly influence the final thesis and any future development of this system.

All footage and events are real. Some may have been generated during real world testing of units and are not fully representative of normal fleet behaviour. All data has been anonymised. Please feel free to message me directly after completing this form if you have any questions or further suggestions.

Prepared for: Cartrack Management

Fleet: 251 Internal Vehicles

Evaluation Period: 5 January 2026 – 10 January 2026
Camera Orientation: Cabin only

What is this and why does it exist?

AI cameras can detect phone use, fatigue, seatbelt violations and more. But they are expensive, the system can be opaque and can be difficult to present to clients meaningfully.

This prototype attempts to solve that by using open-source AI models to review standard dashcam footage and turn any camera into an AI camera. The result is fed into a driver ranking system (Built by you guys, thank you again) and a chatbot that fleet managers can query directly.

In short: Improved rankings, visual evidence, and a natural language interface.

What did it find?

The system reviewed approximately 1.5 million telematics events (Including ranked events such as speeding as well as unranked events such as ignition on) and just over 4,000 dashcam videos.
In reviewing the 4000 dashcam videos. These are the rankable events that the AI model found:
New safety events detected (no prior telematics flag - generally non-AI cameras): 1,670

AI camera events confirmed: 657
AI camera events contradicted: 643
Additional events found on top of existing AI camera detections: 463

Manager Query Examples:

* Indicates required question

1. What is your current role? *

____

2. Years in fleet/telematics industry? *

Mark only one oval.

- 0-2 years
- 3-5 years
- 6-10 years
- 11+ years

3. What fleet type do you work with? *

Mark only one oval.

- Passenger
- Light commercial
- Heavy commercial
- Mixed/Multiple fleet types

4. How many vehicles do you approximately manage? *

Mark only one oval.

- 1-50 vehicles
- 51-200 vehicles
- 201-1000 vehicles
- 1000+ vehicles

5. Example 1: *
The model was asked to "Provide a full safety assessment and performance review of my drivers."

- The model was given 3 random vehicles
- The image shows a snippet of the three vehicles given. As well as some evidence the model provided for the manager.

The models answer summarised:
Vehicle ID 307989723:
Phone Usage: Immediate action is required to address excessive phone use while driving.
Short Speeding: Implement measures to reduce speeding incidents.
Harsh Driving: Monitor and provide training on maintaining a smoother driving style.

Vehicle ID 334006437:
Crash Prevention: Focus on preventing future crashes through additional safety training or reminders about safe driving practices.

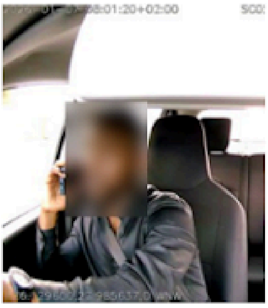
Vehicle ID 509636001:
Drive Hours: Increase drive hours to gather more reliable data for a comprehensive assessment.
Harsh Driving: Monitor and address any harsh driving behaviors as they become evident with increased drive time.

Conclusion:

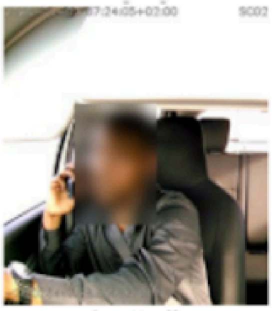
Driver of Vehicle ID **307989723** is the most dangerous due to excessive phone usage, short speeding incidents, and overall high situational risk score. Immediate corrective actions are necessary to mitigate these risks and improve safety standards in your fleet.

Please rank the summarised response from the model below

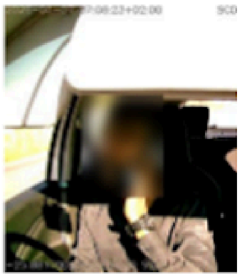
vehicle_id	307989723	334006437	509636001
fleet_rank	1.000	2.000	3.000
ubi_proxy_score	0.680	0.430	0.430
drive_hours	76.330	26.780	8.910
distance_km	960.000	216.000	597.000
crash_count	0.000	1.000	0.000
phone_per_hour	0.052	0.000	0.112
fatigue_per_hour	0.026	0.037	0.000
harsh_per_hour	0.288	0.485	18.414
short_speeding_per_hour	0.026	0.000	0.000
long_speeding_per_hour	0.000	0.000	0.000
distraction_per_hour	0.105	0.149	0.225
seatbelt_per_hour	0.066	0.000	0.225
power_violations	1.000	0.000	0.000
situational_risk	82.000	4.000	0.000
cam_covered_detections	0.000	0.000	0.000
covered_ratio	0.000	0.000	0.000
total_clips	16.000	3.000	8.000



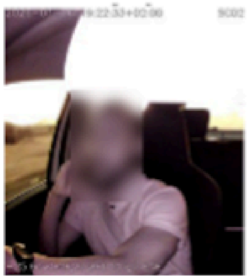
Vehicle 307989723 – 2026-01-07 06:01:21



Vehicle 307989723 – 2026-01-07 05:24:04



Vehicle 307989723 – 2026-01-07 05:06:24



Vehicle 307989723 – 2026-01-06 17:22:32

1	2	3	4	5
☆	☆	☆	☆	☆

6. Do you have any comments on Example 1?

7. Example 2: *
The model was asked to "Which of my drivers are covering their camera?"

- The model was given 100 random vehicles
- The image shows a snippet of the 2 vehicles the model found had camera covered violations. As well as some evidence the model provided for the manager.

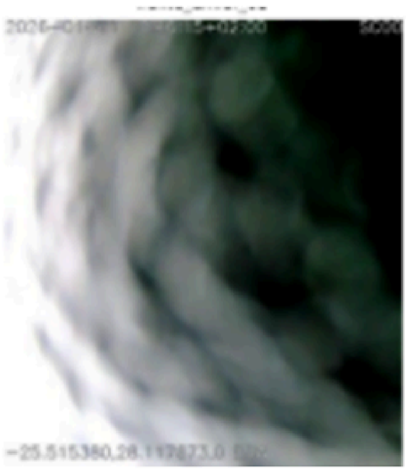
The models answer summarised:
Vehicle 507451763:
Clip: clip_580 (2026-01-11 08:46:13)

Vehicle 312741040:
Clip: clip_4113 (2026-01-05 05:31:55)
Recommendations:

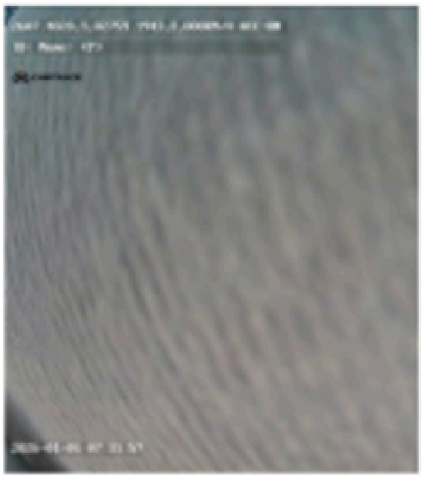
For **Vehicle ID 507451763**, consider a more thorough review of their driving behavior and provide additional training on the importance of camera visibility. Given their high-risk status, deliberate covering could be an issue.

For **Vehicle ID 312741040**, it may be beneficial to remind this driver about the importance of keeping cameras unobstructed during operation. Since there are no crashes and fewer drive hours, a gentle reminder might suffice.

In both cases, ensure that all drivers understand the consequences of covering their cameras and the importance of maintaining clear visibility for safety reasons.



Vehicle 507451763 – 2026-01-11 08:46:13



Vehicle 312741040 – 2026-01-05 05:31:55

1	2	3	4	5
☆	☆	☆	☆	☆

vehicle_id	507451763	312741040
fleet_rank	1.000	2.000
ubi_proxy_score	0.430	0.290
drive_hours	110.260	13.620
distance_km	1019.000	329.000
crash_count	3.000	0.000
phone_per_hour	0.091	0.367
fatigue_per_hour	0.000	0.000
harsh_per_hour	0.045	0.000
short_speeding_per_hour	0.000	0.000
long_speeding_per_hour	0.000	0.000
distraction_per_hour	0.118	0.294
seatbelt_per_hour	0.009	0.661
power_violations	0.000	0.000
situational_risk	8.000	501.000
cam_covered_detections	1.000	1.000
covered_ratio	0.030	0.077
total_clips	33.000	13.000

8. Do you have any comments on Example 2?

9. Thank you for your time and honest feedback. Your input will be referenced directly in the final thesis submission. If anything in this document raised questions or ideas, please reach out.
Feel free to also add final comments below: